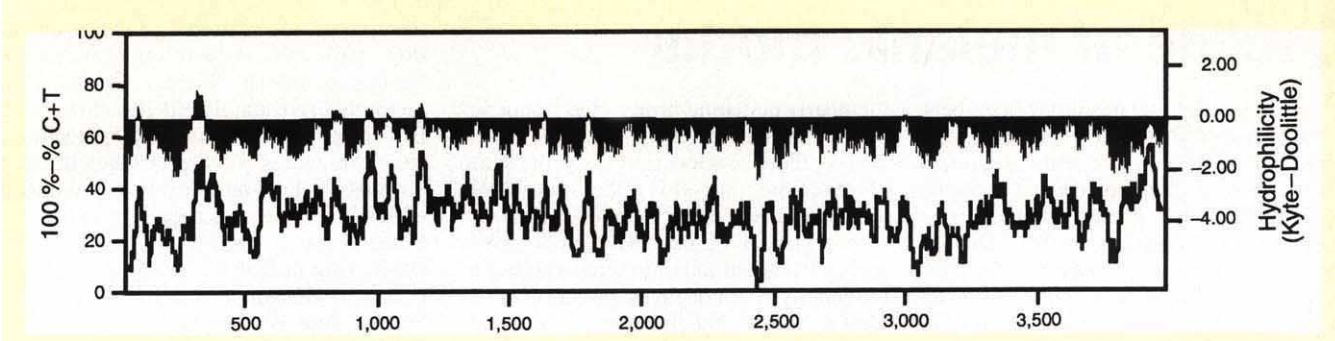

Correspondence between base composition at second positions of codons for the 13 human mitochondrial protein-encoding genes (lower histogram) and the hydrophilicity of corresponding amino acids calculated by the Kyte–Doolittle method (upper histogram) using the MacVector Program (IBI). Base compositions were computed by averaging C and T frequencies using a sliding window of 40 bases for the entire sequence. Similar figures and patterns of base compositional bias at the three codon positions of protein-encoding genes were also determined for rat, mouse, cow, possum, chicken, frog, carp, lancet, sea urchin (2 species) and *Drosophila* (data not shown).

strong bias for or against any of the four bases at the first position in these taxa. At second positions there is a striking compositional bias for C and T. Finally, there is a strong bias for A (or against G) at third positions. The different base compositions probably reflect different constraints operating at each of the codon positions.

We have found that the bias for T and C at second positions is directly correlated with hydrophobicity (see figure). Of the 26 triplets encoding hydrophobic residues, 24 (92%) have T or C at the second position. Mitochondrial proteins contain a high proportion of hydrophobic residues, that are frequently clumped together into domains. These domains probably represent membrane-spanning segments that must be hydrophobic to ensure their conformational stability[8–10]. This functional requirement for hydrophobicity effectively constrains character-state space at second positions to one of two states, T or C, causing second-position sites to cycle between these two states over evolutionary time. Ironically, these positions, hitherto regarded as perhaps the most reliable for inferring evolutionary histories of distantly related taxa, may actually carry less phylogenetic signal and be less reliable than the more fast-evolving first positions whose compositional bias is less skewed.

Here we have identified one constraint that influences mtDNA sequence evolution. If we are to build realistic models of DNA sequence evolution that better allow us to retrieve phylogenetic signal from such data, it will be important to identify others.

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Complex bacterial patterns

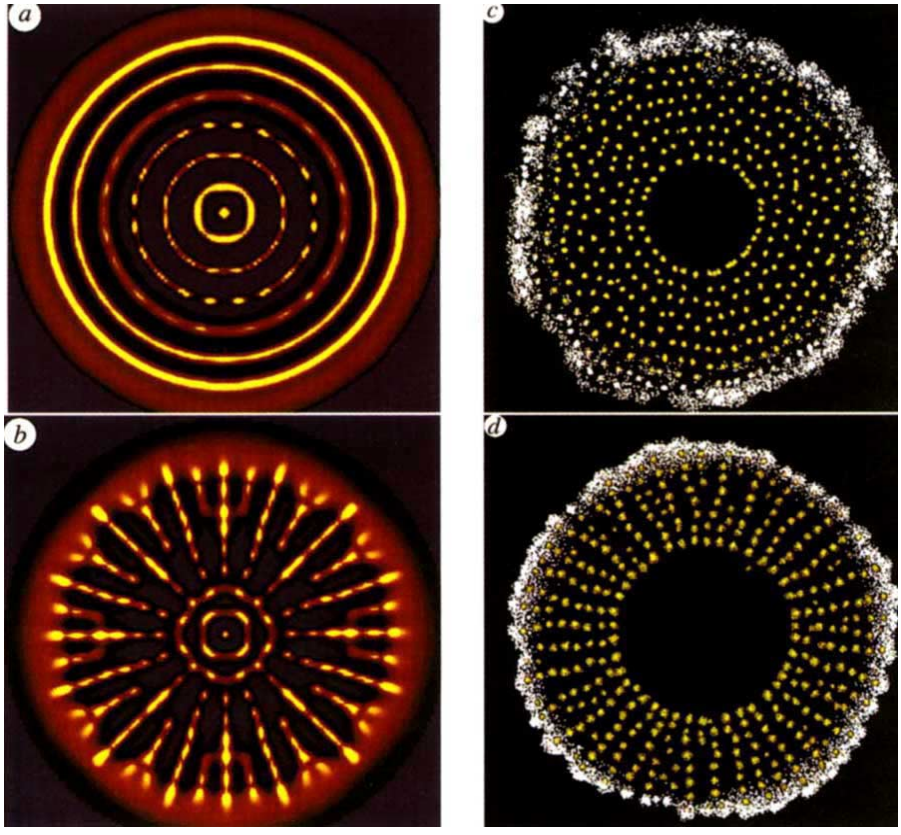
SIR — Budrene and Berg[1] have studied patterns of spots, stripes and rings formed by motile cells of *Escherichia coli* grown in a thin layer of semi-solid agar. Similar patterns are produced by *Salmonella typhimurium* (Y. Blat and M. Eisenbach, personal communication). Here we propose that the interplay of a few common cooperative strategies underlies this diverse set of phenomena. We use two kinds of models to demonstrate that these strategies (the generic features), rather than the details of the models, are crucial for producing the observed patterns.

The most familiar strategy for ensuring cooperative behaviour is attractive chemotaxis. Indeed, Budrene and Berg[1] proposed that a chemoattractant mediates the observed spot formation in *E. coli.* This is easily confirmed by using a model of growth that includes nutrient diffusion and consumption, and bacterial growth and bacterial motion[2]. The addition of a diffusing attractant c that is constantly emitted by the bacteria, together with bacterial motion towards its gradient, leads to the creation of spots. We show results of numerical simulation of a continuous model including these features (*a* in the figure).

This simple model is insufficient, however, to explain several crucial observations[1]. In the experiments, spots appear sequentially in the wake of a spreading broad ring and later 'lock' into position as the bacteria turn non-motile. To capture these effects one must introduce additional mechanisms. First, we explicitly include in the model a 'triggering' field w (a field that must reach a threshold before attractant emission is activated). The value of this threshold may depend on the ambient chemoattractant concentration. In detail, if $c > c_0$, emission occurs when $w > w_0$ and otherwise, when $w > w_1$ (where $w_1 > w_0$). Biologically, w represents the hazardous waste products produced by the bacteria. Also, bacteria in the full model differentiate into a non-motile state when starved for a sufficiently long time. To observe (in the simulations) different patterns, we vary the model's parameters related to the bacterial response to the triggering field and to the precise nature of the chemoattractant signalling. The final continuous model is then in good agreement with many of the features seen in experiment, including the possible radial organization of spots or stripes (*b* in the figure)

An alternative model — the 'communicating walkers' approach[3] (introduced in the study of branching patterns during growth of *Bacillus subtilis*) — leads to the same results. The pattern shown in *c* of the figure was produced by the discrete model described in ref. 3 which we have modified to apply to *E. coli* in semi-solid agar. The 'freezing' of a walker (which stops moving after an interval of time during which not enough food is available) represents the observed differentiation into a non-motile state of the individual bacterium. Augmenting the bacterial motion and nutrient dynamics are the following additional features. First, a 'triggering' field (representing the waste products). The field is produced by the bacteria at a rate proportional to the rate of food consumption and is decomposed by the bacteria at a fixed rate. Second, a chemoattractant field. The emission of the chemoattractant is stimulated when the triggering field exceeds a threshold value or when the concentration of the attractant in the surrounding of the walkers exceeds a minimum value. Once a walker starts to produce attractant, it does so at a fixed rate for an interval of time such that a fixed amount is released.

The walkers approach has its computational advantages and disadvantages relative to the continuum approach. As each walker represents a coarse-graining of about 10^4–10^5 bacteria (numerically, one cannot handle 10^{10} particles, which is the number of bacteria in a colony), it has an



Results of the numerical simulations of the two models. *a*, Spot formation in a continuous model where the bacteria continuously emit chemoattractant. The growth front breaks into rings which subsequently break into spots (colours indicate bacterial density, with purple the lowest density and yellow the highest). *b*, Radial spots pattern in a continuous model which includes the chemoattractant triggering field described in the text. The dynamics of growth consists of a wide outer band which expands, while the spots are created in its wake (same colour code as in a). *c*, Spot formation in the 'walkers' model. The spots are formed in the wake of a spreading circular band (yellow indicates non-motile walkers and white indicates motile walkers). *d*, Observation of a radial structure of spots when the chemorepellent mechanism is included in the walkers model (same colour code as in *c*). (Contact authors for details).

inherently high level of noise. However, it is more flexible in terms of the easier incorporation of complex bacterial response, such as a possible refractory period after emission of chemoattractant or finite emission time. For example, the pattern shown in *c* was produced in the presence of such a finite emission period. In general we found that 'sunflower'-like patterns are formed more easily when finite emission period is included and for high levels of the waste threshold w_0.

As we have seen, radially organized structures can be produced if the triggering mechanism is taken to be attractant-concentration-dependent. In the absence of this autocatalytic effect, neither model could produce the observed radial structures. Motivated by the proposed repulsive chemotaxis in *B. subtilis*[3], we have also considered the possibility that a sunflower-like pattern is changed to a radial structure due to a chemorepellent that is emitted by starved bacteria. We used the walker model to test this possibility and found that chemorepellent signalling can indeed be an alternative explanation. In *d* we show an example of a simulated radial arrangement of spots in the presence of chemorepellent signalling. Other observed patterns such as radial stripes can be obtained if we change the relative strength of the repellent and attractant effects. This result suggests that *E. coli* and *S. typhimurium* might also use a repellent–attractant chemotactic interplay in the cooperative formation of complex patterning. A more detailed comparison with the time evolution of the observed patterns will be needed to distinguish between these hypotheses.

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Biological levels

SIR — As a chemical engineer who has worked in cellular biology, I read with interest your article "Gaps in Molecular Biology" in *Nature*'s 125th anniversary issue[1]. From my perspective, I see a serious limitation in a fundamental guidepost underlying current work in molecular and cellular biology: that a satisfying understanding of complex biological processes will follow from a complete inventory of the molecular components and a knowledge of their structure and reactions (for example, Fig. 6 of ref. 2).

The notion that "knowing all the players" and "who talks to whom" (I. Dawid, cited in ref. 3) will lead to biological enlightenment involves a tacit assumption of linear reactions. Indeed, *Nature* and other journals commonly summarize advances in molecular biology with linear reaction sequences, for example, Ras activates Raf activates MAPK, and so on, or matrix sums of such sequences. But the vast majority of biological reactions are nonlinear (for example, $A + B \rightarrow C$, or $2A \rightarrow B$) and the cell appears as an open system far from equilibrium. In chemical engineering it is understood that as few as four nonlinear reactions in an open system can lead to a spectacular array of limit cycles, chaos, and multiple steady states (see ref. 4). Added complications such as diffusion and concentration gradients lead to unpredictable spatial patterns as well.

Thus one can know all the apparent players, the principal reactions and the rate constants, and yet be unable to predict a specific outcome. The cell is seen to have all the behaviours above and biologists routinely find that some small insults push the cell past a critical point and initiate distant effects, yet linear reasoning persists in most experimental approaches. For example, a common experimental strategy of determining molecular function is to increase or decrease a particular reactant, decreasing expression of a particular gene product, say, and then measuring the consequences. But in complex systems, there may be little or no effect from such interventions because chaos, for example, can provide unexpected stability. The evident surprise of molecular biologists on the organismal effects of many gene 'knockout' experiments may be based on a relative lack of appreciation of nonlinear effects in chemical systems. I find it ironic that the standard wall charts of biochemical reactions show hundreds of coupled nonlinear reactions, while graduate biology students are tacitly encouraged to think in terms of linear cause and effect.